

HEIDELBERG, Germany

WMO: 107345

Lat: 49.393N

Lon: 8.652E

Elev: 358

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 90-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.3	21.5	10.1	9.3	21.1	14.1	11.4	25.3	24.2	47.5	20.9	47.0	4.3	60	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	89.9	68.8	86.3	67.5	82.8	65.7	70.9	84.3	69.3	82.3	67.7	79.8	6.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	98.8	74.9	64.7	92.7	74.0	63.0	87.3	72.1	35.0	84.8	33.6	82.1	32.4	79.8	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	15.9	13.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	36.8	39.1	45.7	52.0	60.4	64.8	70.8	69.8	60.7	52.1	44.0	38.1
	DBStd	13.60	8.22	8.32	6.27	6.88	6.44	6.14	6.47	6.24	5.16	7.14	6.47	7.07
	HDD50	1568	410	309	159	55	4	0	0	0	0	64	194	373
	HDD65	4897	873	725	598	393	170	79	19	27	146	401	631	835
	CDD50	2640	2	4	26	115	325	445	643	613	322	129	13	3
	CDD65	497	0	0	0	4	26	75	197	175	18	2	0	0
	CDH74	4494	0	0	0	52	246	613	1885	1543	148	7	0	0
	CDH80	1521	0	0	0	5	34	151	741	558	32	0	0	0
(14) Wind	WSAvg	5.9	6.6	6.1	6.7	6.5	6.1	5.7	5.6	5.3	5.2	5.1	5.2	6.3
(15) Precipitation	PrecAvg	31.3	2.3	2.2	2.2	2.0	3.0	2.9	3.2	2.6	2.5	2.7	2.8	2.9
	PrecMax	40.9	4.8	5.2	6.2	5.2	6.9	4.8	6.6	8.1	5.1	7.5	5.0	6.5
	PrecMin	21.6	0.2	0.7	0.5	0.1	0.7	0.7	1.0	0.3	0.6	0.6	0.2	0.4
	PrecStd	4.6	1.2	1.2	1.3	1.1	1.5	1.1	1.4	1.7	1.2	1.4	1.2	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.6	61.2	68.3	79.1	84.3	87.7	94.8	95.1	85.6	74.8	60.5	56.8
		MCWB	50.5	51.3	54.7	59.2	64.5	67.5	69.4	69.8	65.1	63.5	53.4	51.8
	2%	DB	53.6	57.0	62.8	74.9	79.2	84.2	91.2	89.8	77.3	69.5	57.2	53.9
		MCWB	49.1	49.2	51.4	57.5	61.9	66.4	70.1	68.7	63.1	60.8	52.0	50.3
	5%	DB	50.4	53.5	59.2	70.1	76.9	80.8	87.9	86.0	73.7	66.1	55.2	51.9
		MCWB	46.5	47.1	49.9	55.4	60.4	64.7	67.9	67.6	61.6	58.6	50.8	47.8
	10%	DB	48.3	50.2	55.7	65.8	73.4	77.1	84.2	82.4	70.2	63.0	53.2	49.7
		MCWB	45.3	45.6	48.6	53.4	58.4	63.4	66.8	65.5	60.1	57.1	49.3	46.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	52.9	56.4	61.3	67.4	71.7	73.4	73.2	67.1	64.0	55.2	53.6
		MCDB	55.5	60.0	65.4	75.4	80.8	80.4	89.1	87.5	77.4	72.1	57.6	55.5
	2%	WB	49.8	50.3	53.3	58.1	63.9	68.3	71.0	70.8	64.9	62.0	52.9	51.4
		MCDB	52.9	55.0	60.3	71.8	75.4	80.1	85.6	83.8	73.5	67.7	56.3	54.0
	5%	WB	47.4	48.3	51.4	55.8	61.9	66.5	69.5	69.3	63.4	60.1	51.2	48.7
		MCDB	50.5	53.0	57.1	67.6	72.9	77.5	83.5	81.7	71.2	64.9	54.4	51.7
	10%	WB	45.2	46.0	49.6	54.1	60.2	64.7	67.9	67.5	61.6	58.0	49.6	46.1
		MCDB	47.8	49.6	55.5	64.7	70.2	75.1	80.8	78.9	69.0	62.4	52.2	49.0
(35) Mean Daily Temperature Range	MDBR		9.1	11.4	13.8	17.1	18.4	16.8	18.3	18.9	16.5	13.8	9.4	7.8
	5% DB	MCDBR	11.0	17.6	20.7	24.1	25.1	23.2	24.5	25.3	23.2	18.7	12.0	10.0
		MCWBR	8.8	11.4	12.0	11.1	11.4	9.9	8.6	9.7	11.2	11.2	8.4	8.1
	5% WB	MCDBR	10.8	15.1	18.0	22.4	21.8	20.7	22.0	21.9	18.6	16.2	11.1	9.7
		MCWBR	9.4	10.4	11.3	10.9	10.6	9.5	8.5	9.5	9.9	10.2	8.6	8.2
(40) Clear-Sky Solar Irradiance	taub		0.327	0.355	0.388	0.418	0.426	0.429	0.431	0.426	0.397	0.379	0.350	0.324
	taud		2.387	2.325	2.246	2.203	2.215	2.241	2.230	2.269	2.323	2.358	2.381	2.384
	Ebn at Noon		227	247	257	262	265	264	262	257	252	235	214	210
	Edh at Noon		21	28	36	42	43	42	42	39	33	27	21	19
(44) All-Sky Solar Radiation	RadAvg		269	513	885	1363	1617	1808	1706	1478	1085	639	312	212
	RadStd		41	82	128	178	175	146	169	138	130	66	33	29

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page